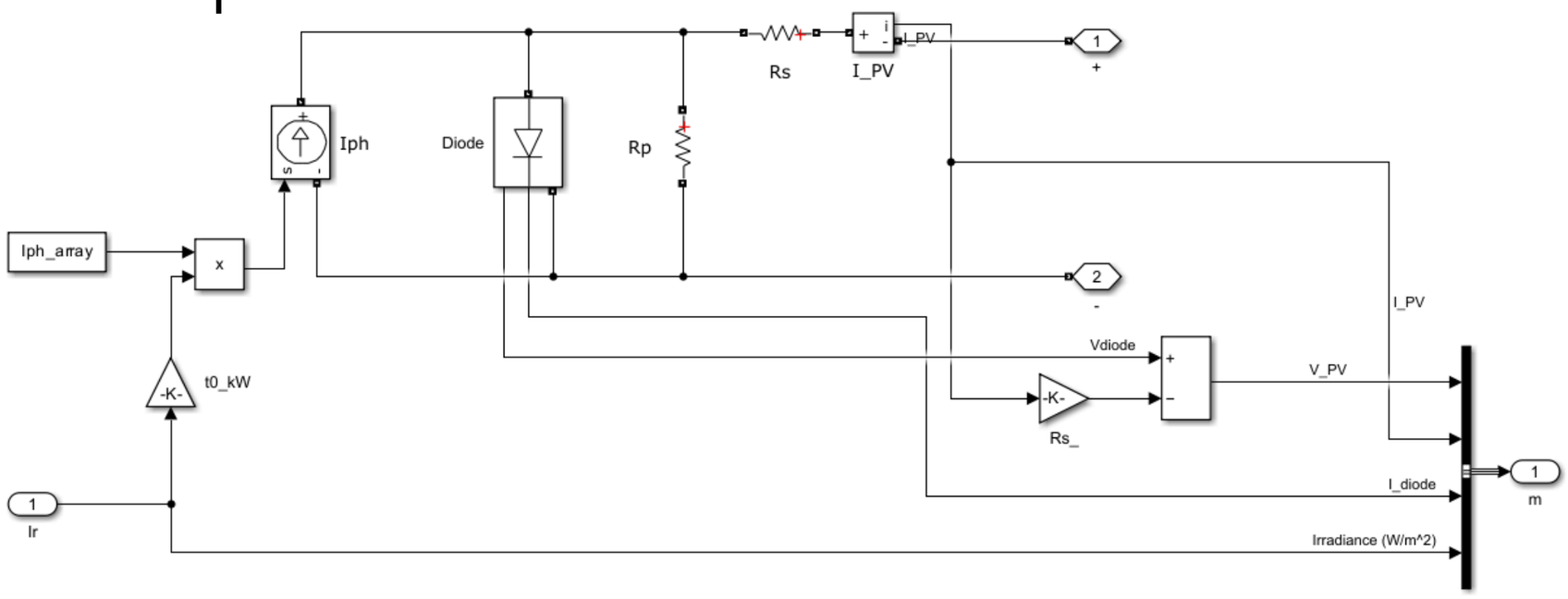
SPV INTEGRATION INTO THE STANDARD iEEE 14 BUS SYSTEM.

Solar panel Equivalent model



Photovoltaic (PV) Array Mathematical Model

The photovoltaic (PV) array is modeled using the single-diode equivalent circuit, which accurately represents the electrical behavior of a practical solar cell while maintaining computational simplicity. The model consists of a photocurrent source **Iph**, a diode, a series resistance **Rs**, and a shunt resistance **Rp**. The photocurrent source represents the current generated due to incident solar radiation, the diode models the p-n junction characteristics of the solar cell, the series resistance accounts for internal resistive losses in the semiconductor material and metallic contacts, and the shunt resistance represents leakage currents caused by imperfections within the cell. Together, these components reproduce the nonlinear current-voltage (I-V) characteristics of a real photovoltaic module.

The primary input to the model is the solar irradiance (G), measured in W/m². When sunlight strikes the surface of the PV cell, photons transfer their energy to electrons in the semiconductor material, creating electron-hole pairs. The internal electric field present at the p-n junction separates these charge carriers, generating a photocurrent. Since the number of incident photons increases with irradiance, the generated photocurrent is approximately proportional to the solar irradiance and can be expressed as

$$Iph = Iph,ref \times \frac{G}{Gref}$$

where ***Iph,ref*** is the photocurrent under standard test conditions (STC), (G) is the actual irradiance, and G*ref*=1000 W/m^2. Therefore, an increase in irradiance primarily increases the output current of the PV module, while the output voltage changes only slightly.

The second important input to the PV model is the cell temperature T. Temperature influences the semiconductor properties and primarily affects the diode characteristics. The thermal voltage of the diode is given by

$$V_T = \frac{KT}{q} \times Qd \times Ncell \times Nser$$

where $K = 1.3806 \times 10^{-23}$ J/K is the Boltzmann constant, $q = 1.6022 \times 10^{-19}$ C is the electronic charge, Qd is the diode ideality factor, Ncell is the number of series-connected cells in a PV module, and Nser is the number of series-connected modules in a string.

As the cell temperature increases, both the thermal voltage and the diode saturation current increase, resulting in a reduction in the open-circuit voltage of the PV module. Consequently, higher operating temperatures decrease the maximum output power and reduce the overall efficiency of the solar panel. The diode in the equivalent circuit represents the nonlinear behavior of the p–n junction. As the terminal voltage increases, the diode current rises exponentially in accordance with the Shockley diode equation.

$$I_D = I_{sat}[exp(\frac{V_D}{V_T}) - 1]$$

where Id is the diode current, Isat is the diode saturation current, Vd is the voltage across the diode, and VT is the thermal voltage. At low terminal voltages, the diode current is negligible and most of the generated photocurrent is delivered to the external load. However, as the terminal voltage approaches the open-circuit voltage, the diode current increases rapidly and opposes the photocurrent, thereby limiting the output current of the PV module.

The series resistance Rs represents the internal resistance of the solar cell, including the resistance of the semiconductor material, metal contacts, and electrical interconnections. As the output current flows through this resistance, a voltage drop occurs according to Ohm's law,

$$V_{drop} = I_{PV} \times R_S$$

This voltage drop reduces the terminal voltage available at the output of the PV module. Therefore, the terminal voltage is obtained by subtracting the series voltage drop from the internal diode voltage,

$$V_{PV} = V_{diode} - I_{PV} \times R_s$$

SPV INTEGRATION INTO THE STANDARD iEEE 14 BUS SYSTEM.

A higher value of series resistance (Rs) increases conduction losses, reduces the fill factor, and decreases the maximum power output of the PV array.

The shunt resistance (Rp) represents the leakage current paths caused by manufacturing defects and impurities in the semiconductor material. The leakage current flowing through the shunt resistance is given by:

$$I_{sh} = \frac{R_{PV} + I_{PV} R_S}{R_{PV}}$$

Ideally, the shunt resistance should be infinitely large so that no current bypasses the external load. However, a decrease in shunt resistance results in increased leakage current, thereby reducing the output current and overall efficiency of the PV module.

Applying Kirchhoff's Current Law (KCL) to the equivalent circuit, the output current supplied by the photovoltaic module is obtained as the difference between the generated photocurrent, the diode current, and the shunt leakage current:

$$I_{PV} = I_{ph} - I_D - I_{sh}$$

Substituting the expressions for the diode current and shunt current yields the complete single-diode PV model equation:

$$I_{PV} = I_{ph} - I_{sat} \left[exp\left(\frac{V + V_{PV} R_S}{V_T} \right) - 1 \right] - \frac{V + V_{PV} R_S}{R_P}$$

This nonlinear equation governs the electrical behavior of the photovoltaic array and is solved numerically during simulation to obtain the output voltage and current for different operating conditions.

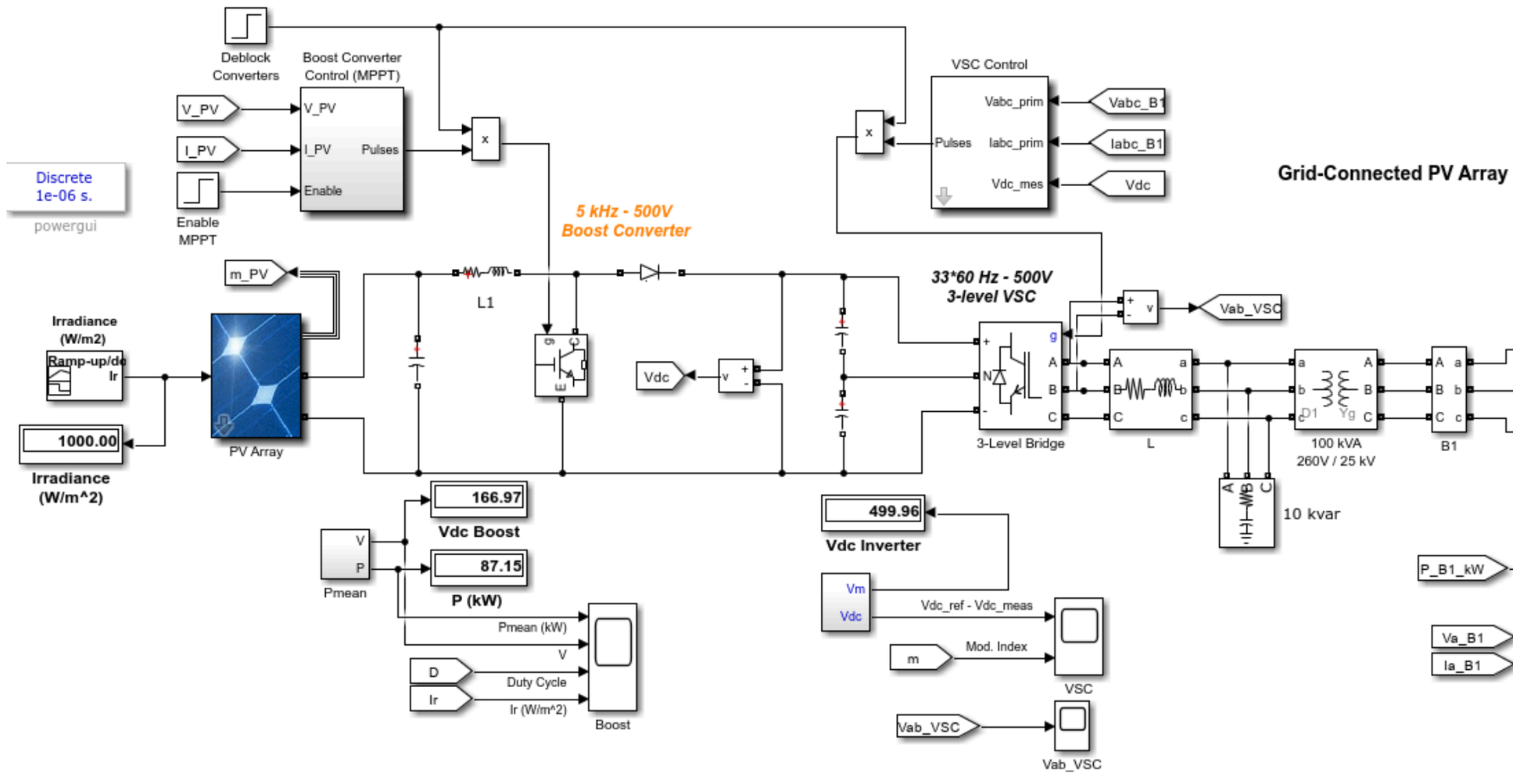
In the Simulink implementation shown, the irradiance input is first converted into the corresponding photocurrent, which drives the controlled current source representing the generated solar current. Simultaneously, the temperature input is used to calculate the thermal voltage required in the diode equation. The diode block computes the exponential junction current, while the shunt resistance accounts for leakage losses. The output current is measured using the current sensor, and the voltage drop across the series resistance is calculated and subtracted from the internal diode voltage to obtain the terminal voltage of the PV array. Finally, the model outputs the terminal voltage Vpv, output current Ipv, diode current, and irradiance, which are subsequently supplied to the MPPT controller and the DC-DC boost converter for maximum power extraction and grid integration.

DC–DC Boost Converter

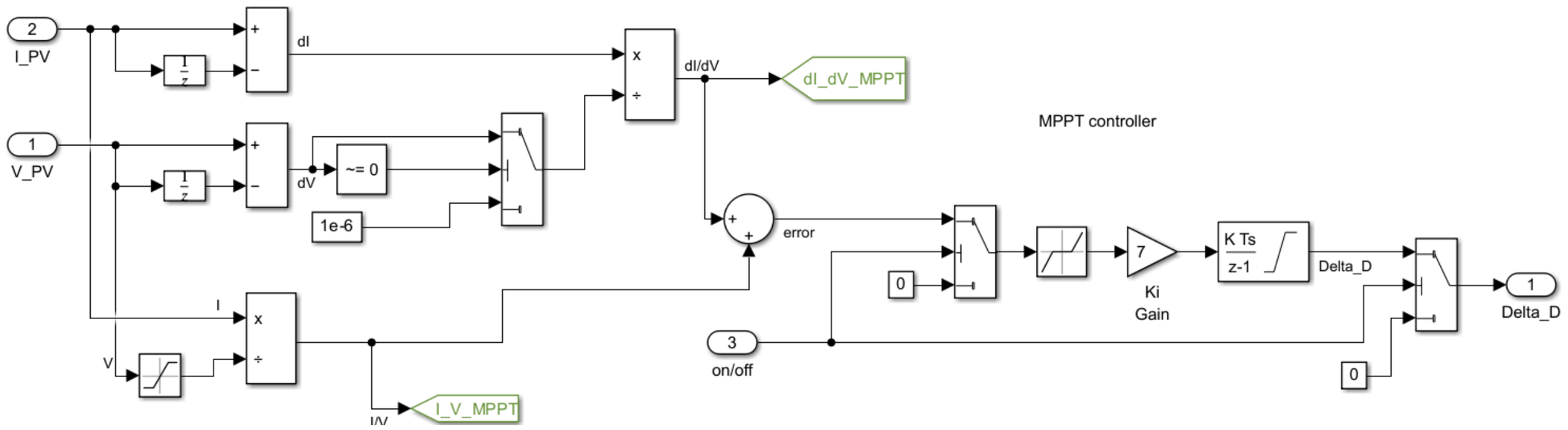
A DC–DC boost converter is employed to step up the variable output voltage of the PV array to the required DC-link voltage for the grid-connected inverter. In this system, the PV array output of approximately **272 V** at the maximum power point is boosted to a regulated **500 V DC**. The converter operates at a switching frequency of 5 kHz, and its output voltage is controlled by varying the duty cycle according to the MPPT algorithm. The voltage gain of an ideal boost converter operating in continuous conduction mode is given by:

$$V_o = \frac{V_{in}}{1 - D}$$

where Vo is the output voltage, Vin is the input voltage, and D is the duty cycle. The boost converter **ensures a stable 500 V DC-link** voltage, enabling efficient power transfer from the PV array to the inverter.



SPV INTEGRATION INTO THE STANDARD iEEE 14 BUS SYSTEM.



Maximum Power Point Tracking (MPPT) is a control technique that continuously adjusts the operating point of a photovoltaic (PV) array to ensure it operates at its Maximum Power Point (MPP), where the maximum available power is extracted

.

MPPT is essential because the MPP varies with changes in solar irradiance and temperature. By continuously tracking the MPP, it maximizes energy extraction and improves the overall efficiency of the PV system.

3.X MPPT Controller Using Incremental Conductance Method

The Maximum Power Point Tracking (MPPT) controller employed in this model is based on the Incremental Conductance algorithm combined with a discrete integral regulator to ensure accurate and stable tracking of the maximum power point (MPP) under varying irradiance and temperature conditions. The controller continuously measures the photovoltaic (PV) array voltage Vpv and current Ipv, and determines the operating point relative to the MPP by evaluating the slope of the PV power-voltage characteristic.

The output power of the PV array is expressed as

$$P = VI$$

Differentiating the power with respect to voltage gives:

$$\frac{dP}{dV} = I + V\frac{dI}{dV}$$

At the maximum power point,

$$\frac{dP}{dV} = 0$$

which results in

$$I + V\frac{dI}{dV} = 0$$

or equivalently,

$$\boxed{\frac{dI}{dV} = -\frac{I}{V}}$$

At the MPP, this error becomes zero. A positive error indicates that the operating point lies to the left of the Maximum Power Point (MPP), whereas a negative error indicates operation to the right of the MPP. The objective of the controller is therefore to continuously drive the error towards zero.

To improve numerical stability, a small value (1×10^{-6}) is substituted whenever ΔV approaches zero, thereby preventing division by zero during the calculation of $\Delta V/\Delta I$. Furthermore, a Dead Zone block is incorporated to suppress very small error values arising from measurement noise and switching ripple. This prevents unnecessary duty cycle oscillations when the operating point is already close to the Maximum Power Point.

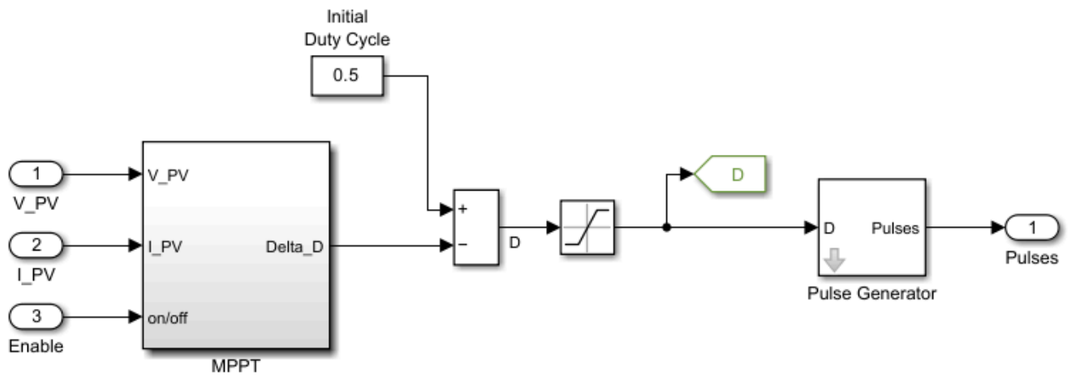
The processed error is then multiplied by the integral gain (K_i) and passed through a discrete-time integral regulator, which generates the duty cycle correction (ΔD):

$$\Delta D = K_i \int e\, dt$$

The final duty cycle supplied to the PWM generator is obtained by modifying the initial duty cycle according to

$$\boxed{D = D_{\text{initial}} - \Delta D}$$

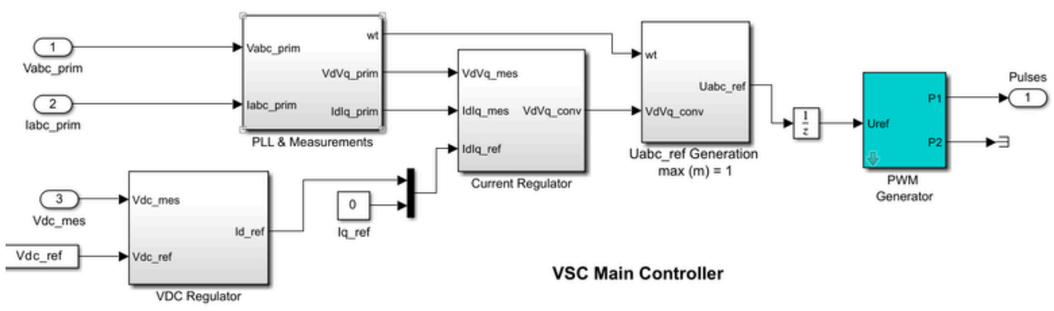
where D_initial = 0.5 is the initial operating duty ratio. A saturation block ensures that the duty cycle remains within its permissible limits (0–1). Finally, the PWM generator converts the duty cycle into switching pulses for the boost converter operating at 5 kHz, thereby regulating the PV array voltage until the operating point converges to the maximum power point. An enable/deblock signal is also incorporated to activate or disable the MPPT controller and converter operation, providing safe startup and shutdown functionality.



SPV INTEGRATION INTO THE STANDARD iEEE 14 BUS SYSTEM.

Three-Level Voltage Source Converter (VSC)

The **500 V DC** output from the boost converter is supplied to a **three-level Voltage Source Converter (VSC**), which converts it into a **60 Hz three-phase AC** output for grid integration. Two series-connected DC-link capacitors split the input voltage equally, creating a neutral point required for three-level operation while stabilizing the DC-link voltage. The inverter generates three voltage levels +Vdc/2, 0, -Vdc/2, reducing harmonic distortion and improving output waveform quality. An L-filter is used at the inverter output to suppress switching harmonics before power is delivered to the grid.



VSC Main Controller

abc–dq Transformation

To simplify the control of the grid-connected inverter, the measured three-phase voltages and currents are transformed from the stationary abc reference frame to the synchronous rotating dq reference frame using the Park transformation. As shown in the Simulink model, the three-phase grid voltage (Vabc) is first converted into per-unit values and applied to a Phase-Locked Loop (PLL). The PLL synchronizes with the grid voltage and continuously estimates the grid angle (ωt), from which the reference signals sin(ωt) and cos(ωt) are generated.

Using trigonometric relationships, the model generates the phase-shifted reference signals sin(ωt − 2π/3), sin(ωt + 2π/3), cos(ωt − 2π/3), and cos(ωt + 2π/3). These signals are then multiplied with the corresponding phase voltages (or currents), and the products are summed and scaled to obtain the d-axis, q-axis, and zero-sequence components according to

$$V_d = \frac{2}{3}\left[V_a \sin(\omega t) + V_b \sin\left(\omega t - \frac{2\pi}{3}\right) + V_c \sin\left(\omega t + \frac{2\pi}{3}\right)\right]$$

$$V_q = \frac{2}{3}\left[V_a \cos(\omega t) + V_b \cos\left(\omega t - \frac{2\pi}{3}\right) + V_c \cos\left(\omega t + \frac{2\pi}{3}\right)\right]$$

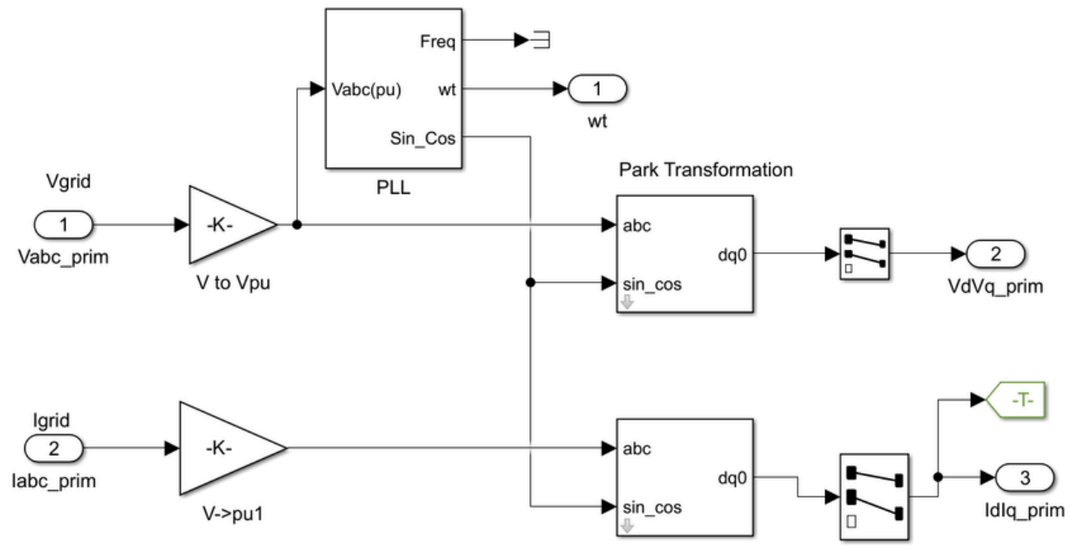
$$V_0 = \frac{1}{3}(V_a + V_b + V_c)$$

Similarly, the three-phase grid currents are transformed into the synchronous reference frame as

$$I_d = \frac{2}{3}\left[I_a \sin(\omega t) + I_b \sin\left(\omega t - \frac{2\pi}{3}\right) + I_c \sin\left(\omega t + \frac{2\pi}{3}\right)\right]$$

$$I_q = \frac{2}{3}\left[I_a \cos(\omega t) + I_b \cos\left(\omega t - \frac{2\pi}{3}\right) + I_c \cos\left(\omega t + \frac{2\pi}{3}\right)\right]$$

$$I_0 = \frac{1}{3}(I_a + I_b + I_c)$$



In the implemented controller, the zero-sequence component is discarded using a Selector block because the grid-connected three-phase system operates under balanced conditions. Consequently, only the d-axis and q-axis components are supplied to the control loops. Since the PLL aligns the rotating reference frame with the grid voltage vector, the transformed d and q quantities become nearly constant under steady-state conditions. This conversion of sinusoidal three-phase signals into DC quantities significantly simplifies PI controller design and enables independent regulation of active and reactive power in the grid-connected inverter.

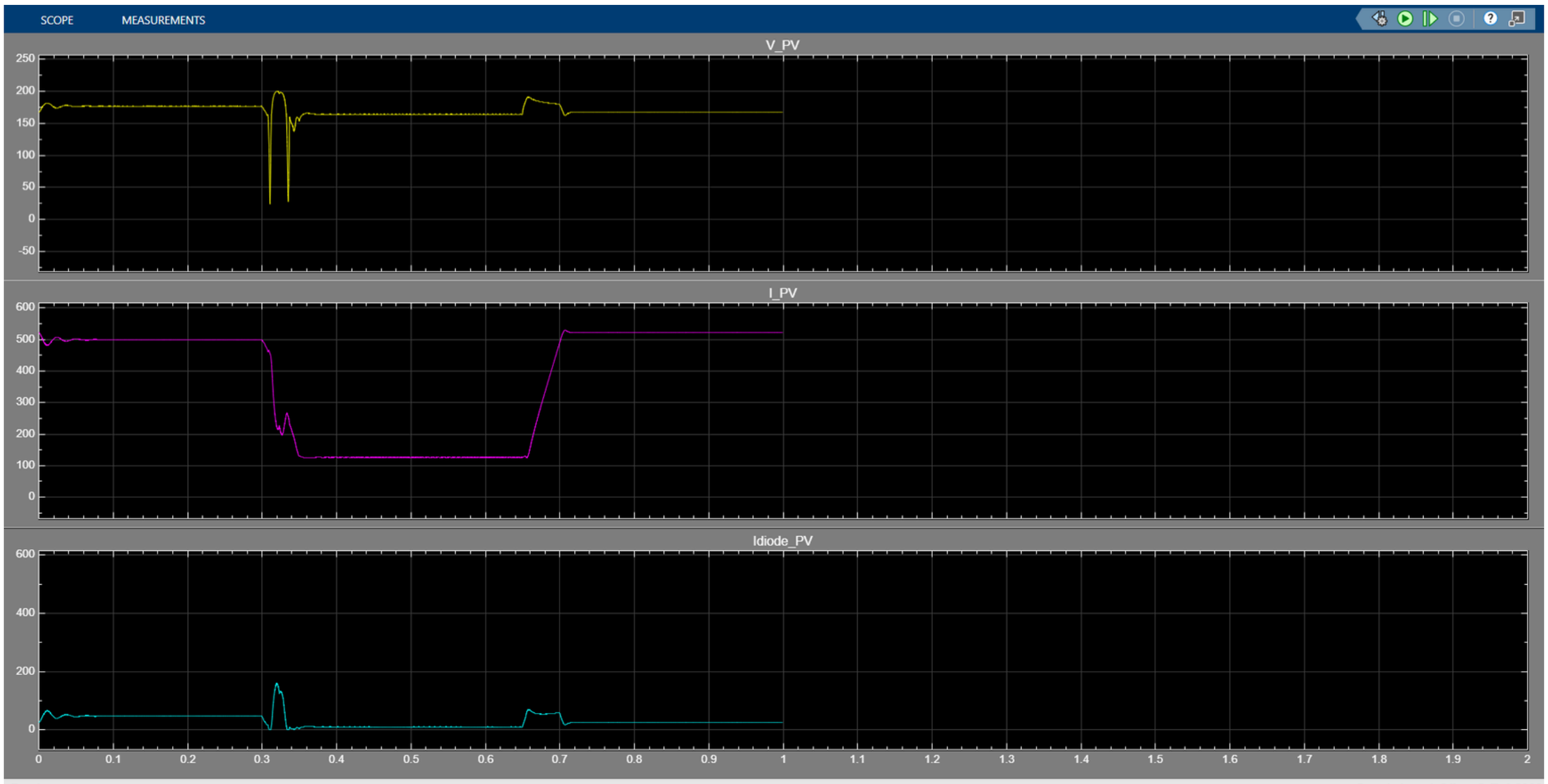
The measured DC-link voltage is compared with its reference value in the VDC Regulator, which generates the reference d-axis current **Id** required to maintain a constant DC-link voltage. The q-axis current reference **Iq** is set to zero to achieve unity power factor operation.

The Current Regulator compares the reference currents with the measured currents and calculates the required converter voltage components in the dq reference frame. These voltage references are then converted back into three-phase reference voltages by the Uabc_ref Generation block using the grid angle obtained from the PLL.

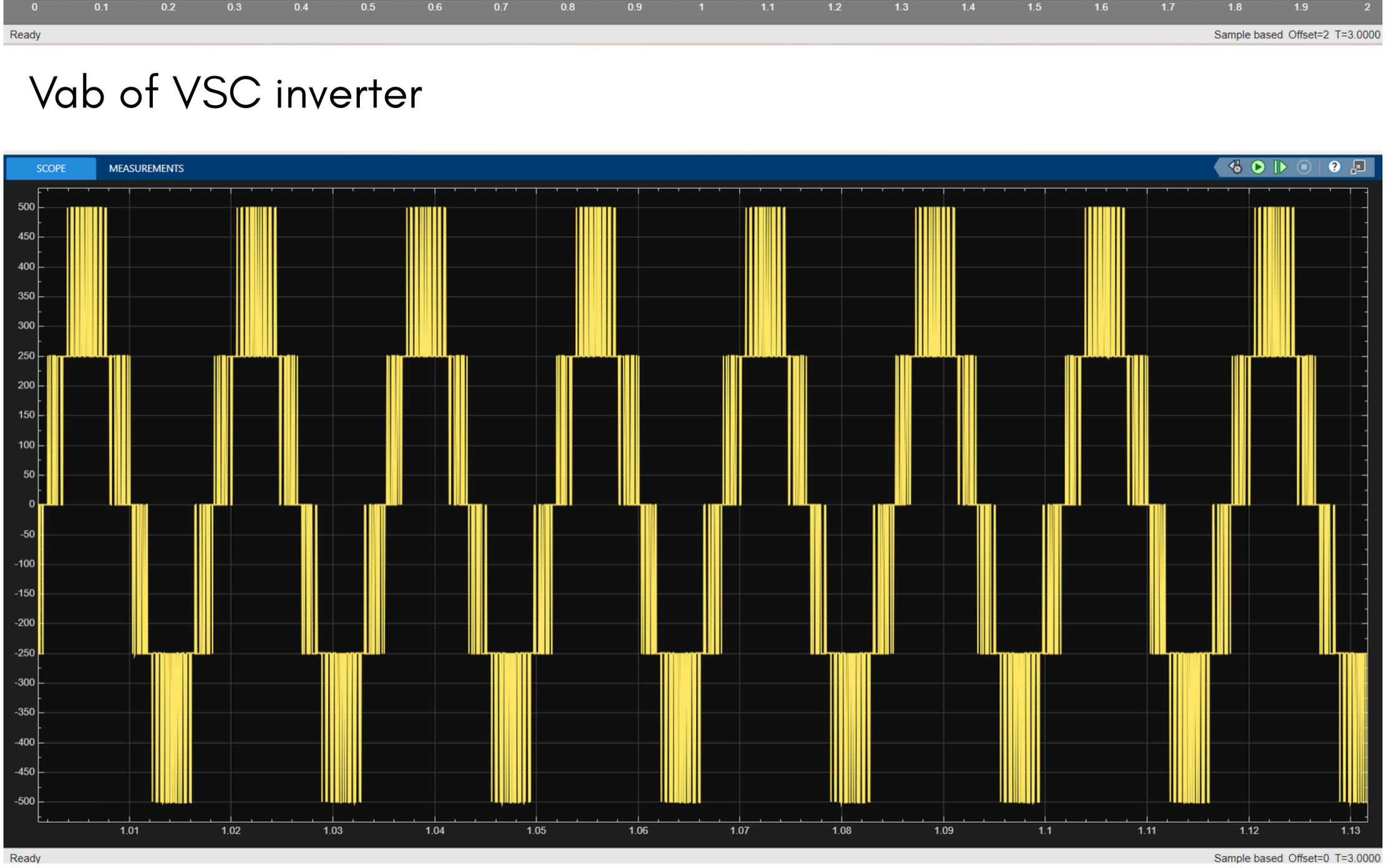
Finally, the PWM Generator converts the three-phase reference voltages into switching pulses, which are applied to the Voltage Source Converter (VSC). These pulses control the inverter to inject the desired power into the grid while maintaining the DC-link voltage at its reference value.

SPV INTEGRATION INTO THE STANDARD iEEE 14 BUS SYSTEM.

PV ARRAY VOLTAGE, CURRENT, DIODE CURRENT

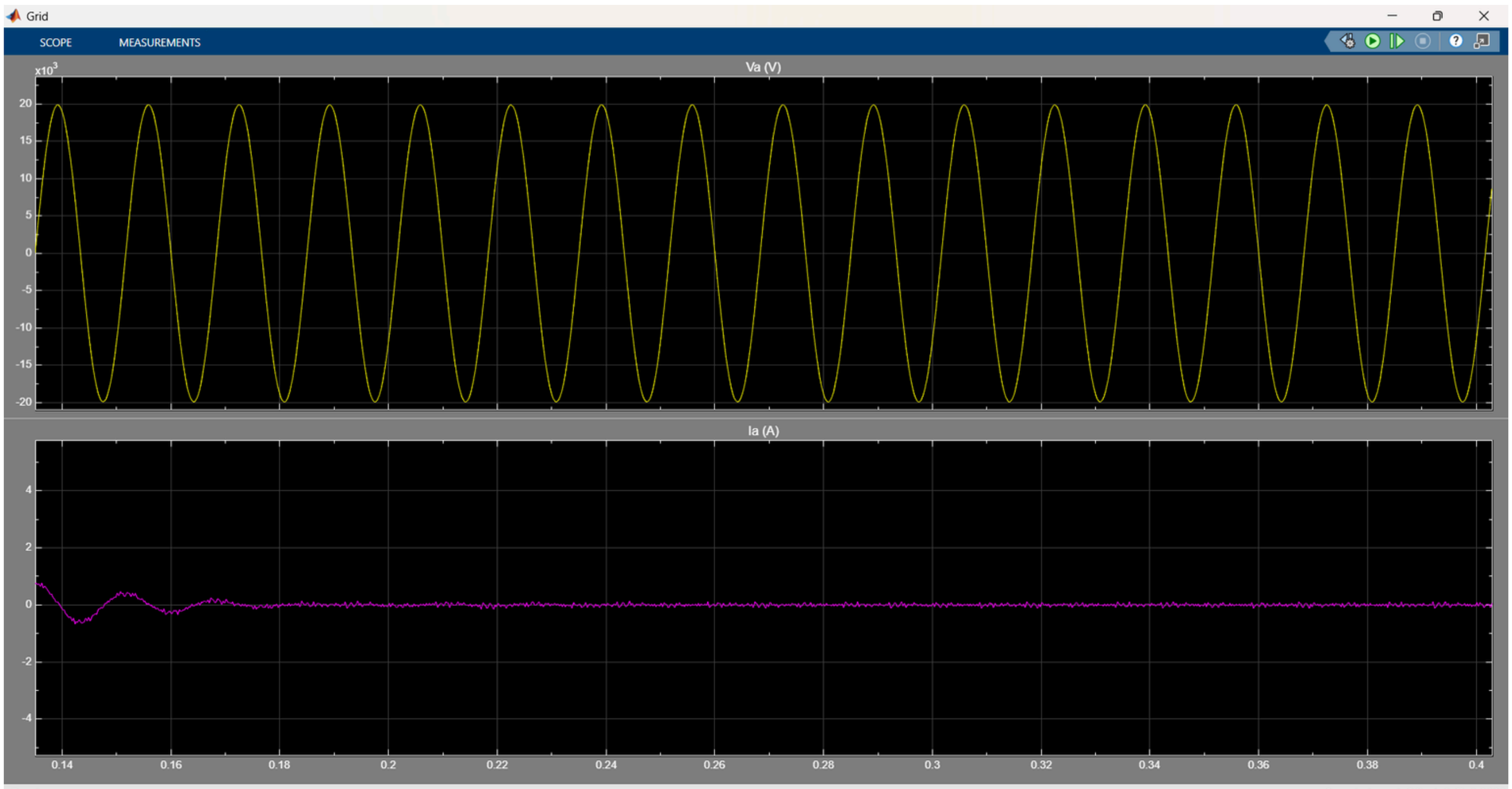


Vab of VSC inverter

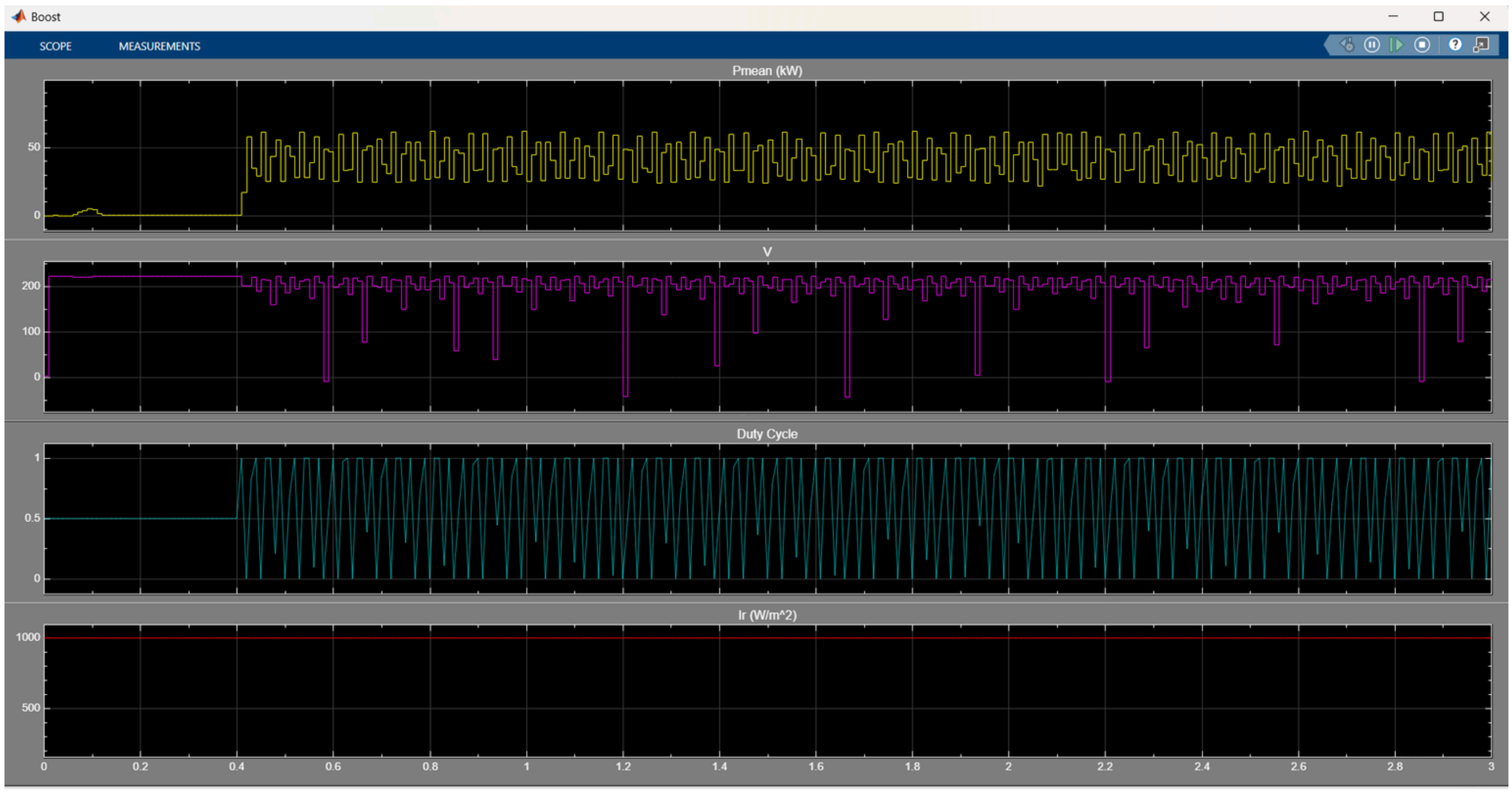


SPV INTEGRATION INTO THE STANDARD iEEE 14 BUS SYSTEM.

Grid voltage



Boost converter output



3 MW SPV series and parallel configuration

